

Theorem (Triangle Inequality). *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner-product space and let $x, y \in V$. Then*

$$\|x + y\| \leq \|x\| + \|y\|,$$

where the norm is defined by $\|v\| = \sqrt{\langle v, v \rangle}$ for every $v \in V$.

Proof. **Norm definition.** For any $v \in V$,

$$\|v\| := \sqrt{\langle v, v \rangle} \geq 0,$$

since $\langle v, v \rangle$ is a non-negative real number.

Zero-vector cases. If $x = 0$ then $\|x + y\| = \|y\| = \|x\| + \|y\|$; similarly if $y = 0$. Henceforth assume $x \neq 0$ and $y \neq 0$.

Expansion of $\|x + y\|^2$. Using linearity in the first argument and conjugate symmetry,

$$\begin{aligned} \langle x + y, x + y \rangle &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + \|y\|^2 + (\langle x, y \rangle + \overline{\langle x, y \rangle}) \\ &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2. \end{aligned}$$

Consequently,

$$\|x + y\|^2 = \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2. \quad (1)$$

Cauchy–Schwarz inequality. For any vectors x, y in an inner-product space,

$$|\langle x, y \rangle| \leq \|x\| \|y\|, \quad (2)$$

with equality iff x and y are linearly dependent. Since $\operatorname{Re} z \leq |z|$ for any complex number z , we obtain

$$\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle| \leq \|x\| \|y\|. \quad (3)$$

Upper bound for $\|x + y\|^2$. Substituting (3) into (1) yields

$$\|x + y\|^2 \leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 \quad (1)$$

$$= (\|x\| + \|y\|)^2. \quad (4)$$

Taking square roots. Both sides of (4) are non-negative, and the square-root function is monotone on $[0, \infty)$; therefore

$$\|x + y\| \leq \|x\| + \|y\|.$$

Equality condition. Equality in the triangle inequality occurs precisely when equality holds in (4). Equality in Cauchy–Schwarz requires $y = \lambda x$ for some $\lambda \in \mathbb{C}$; equality in $\operatorname{Re} z \leq |z|$ forces $\langle x, y \rangle$ to be a non-negative real number, i.e. $\lambda \geq 0$. Thus equality holds iff either one of the vectors is zero or there exists $\lambda \geq 0$ such that $y = \lambda x$. $\square$